# Lecture #17 - Orthogonal Matrices & Gram-Schmidt

3/14/26

orthonormal vectors

$$q_i^T q_j = \begin{cases} 0 & \text{if } i \neq j \\ 1 & \text{if } i = j \end{cases}$$

Matrix $Q = \begin{bmatrix} q_1 \cdots q_n \end{bmatrix}$ → look at $Q^TQ$

$$Q^TQ = \underbrace{\begin{bmatrix} q_1^T \\ q_n^T \end{bmatrix}}_{Q^T} \underbrace{\begin{bmatrix} q_1 \cdots q_n \end{bmatrix}}_{Q} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I}$$

In the case where the matrix is square,
that's when we call it orthogonal
↳ because it has an inverse

If $Q$ is square then $Q^TQ = I$ tells us $Q^T = Q^{-1}$

Ex. permutation $Q = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$ ↗ $QQ^T = I$

↳ $Q^T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

Hadamard Matrix $Q = \frac{1}{2}\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}$ ↗ orthogonal matrix with 1 and -1

$$Q = \frac{1}{3}\begin{bmatrix} 1 & -2 & 2 \\ 2 & -1 & -2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$Q = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

Square roots is very common during Gram-Schmidt because we always want unit vectors (÷ by length)

What's the good of having a Q?

Suppose Q has orthonormal columns.

Project onto its column space

↳ Then what is the projection matrix?

$P = A(A^TA)^{-1}A^T \rightarrow P = Q(Q^TQ)^{-1}Q^T$

$= \underline{\underline{QQ^T}}$

↳ a projection matrix

↳ properties: $\{QQ^T = I$ if Q is square$\}$

↳ a projection is symmetric

↳ for projections - $(QQ^T)(QQ^T) = QQ^T$

projecting $2x = 1x$

orthonormal matrix basis

↳

messy eqns become trivial

the normal eqn: $\underbrace{A^TA}\hat{x} = A^Tb$

↓

now A is Q $\rightarrow \underbrace{Q^TQ}_{I}\hat{x} = Q^Tb$

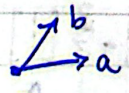
$\hat{x} = Q^Tb$

now $\hat{x}_i = q_i^Tb$

Now let's start with normal vectors and make them orthonormal

Gram-Schmidt

start with independent vectors a, b




and the goal is orthogonal vectors A, B

↳ then orthonormal $q_1 = \frac{A}{\|A\|}$

$q_2 = \frac{B}{\|B\|}$

let's take A (we're happy)

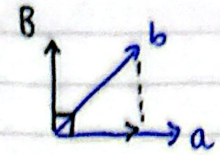
and now B (not happy bc not ⊥ A)

So the idea is to find a vector that starts with b but makes it ⊥ to A

↳

the error vector ("e")

**The IDEA** :




Step 1 : $a = A$

Step 2 : $B = b - \frac{A^T b}{A^T A} A$

Check that $A^T B = A^T \left( b - \frac{A^T b}{A^T A} A \right) = 0$

Shadow of $A = xA$

$b = xA + B$

$B = b - xA$

$\rightarrow$ $B \perp A$ $\rightarrow$ which means dot product $= 0$

$\downarrow$

$A^T B = 0$

$A^T (b - xA) = 0$

$A^T b - x(A^T A) = 0$

$x = \frac{A^T b}{A^T A}$

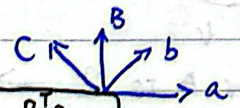
Now add in vector C

$\downarrow$

$q^3 = \frac{c}{\|c\|}$

$\downarrow$

Find vector C that is $\perp$ to A and B




$$\boxed{C = c - \underbrace{\frac{A^T c}{A^T A} A}_{-A} - \underbrace{\frac{B^T c}{B^T B} B}_{-B}} \rightarrow \begin{array}{l} C \perp A \\ C \perp B \end{array}$$

from c, we subtracted its component in A & B direction

ex.

$a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ $b = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$

$\parallel$

$A$

$\frac{A^T b}{A^T A} = \frac{[1\,1\,1]\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}}{[1\,1\,1]\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}} = \frac{3}{3}$

$\rightarrow B = \underset{b}{\begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}} - \frac{3}{3} \underset{A}{\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}} = \underset{B}{\begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}}$ $\rightarrow$ $Q = [q_1 q_2] = \begin{bmatrix} 1/\sqrt{3} & 0 \\ 1/\sqrt{3} & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{2} \end{bmatrix}$

length $A = (1^2 + 1^2 + 1^2) = 3$

$\hookrightarrow$ take $\sqrt{3}$ = length,

so divide A by $\sqrt{3}$

takeaway : $A = QR$

$\downarrow$

magic formula,

expression of Gram-Schmidt